

PROJECT REPORT 1

Basic Vulnerability Assessment for a Small Business Network

Prepared By

Md Washim Baksh

Domain

Cybersecurity | Vulnerability Assessment and Risk Management

Duration

4 Weeks

Abstract

The objective of this project was to perform a comprehensive vulnerability assessment of a simulated small business network infrastructure. The assessment aimed to identify security weaknesses, evaluate potential risks, and recommend mitigation strategies to strengthen the organization's security posture.

A virtual cybersecurity lab environment was established using VirtualBox/VMware. Multiple vulnerable systems, including Metasploitable and OWASP Juice Shop, were deployed to simulate real-world business assets. Industry-standard tools such as Nmap and OpenVAS were used to conduct network discovery, port scanning, and vulnerability assessments.

The findings were analyzed using the Common Vulnerability Scoring System (CVSS) framework to prioritize risks based on severity. Actionable recommendations were then proposed to reduce the attack surface and improve overall security.

1. Introduction

Organizations of all sizes face increasing cybersecurity threats from attackers seeking to exploit vulnerabilities in networks, applications, and systems. Small businesses are particularly vulnerable due to limited security resources and lack of dedicated security teams.

Vulnerability Assessment is a proactive security process that helps organizations identify, analyze, and remediate security weaknesses before they can be exploited by malicious actors.

This project simulates a real-world vulnerability assessment engagement for a small business network and demonstrates the practical application of vulnerability management methodologies.

2. Project Objectives

The primary objectives of this project were:

- Establish a virtual penetration testing environment.
 - Perform network reconnaissance and asset discovery.
 - Identify open ports and exposed services.
 - Conduct automated vulnerability assessments.
 - Analyze discovered vulnerabilities.
 - Prioritize risks using CVSS scoring.
 - Recommend remediation strategies.
 - Improve the overall security posture of the simulated organization.
-

3. Lab Environment Setup

Virtualization Platform

- VirtualBox / VMware Workstation

Operating Systems and Targets

Attacker Machine

- Kali Linux

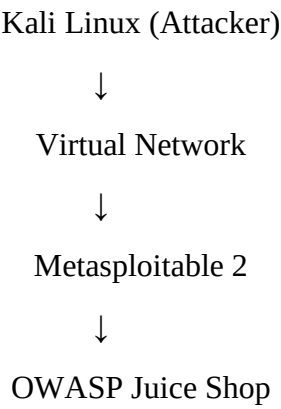
Vulnerable Targets

- Metasploitable 2
- OWASP Juice Shop

Network Configuration

The systems were configured within an isolated virtual network to ensure safe testing and prevent unintended impact on production systems.

Lab Architecture



4. Tools Used

Tool	Purpose
Kali Linux	Security Testing Platform
Nmap	Network Discovery and Port Scanning
OpenVAS (Greenbone)	Vulnerability Assessment
CVSS Calculator	Risk Prioritization
Firefox Browser	Accessing Web Applications
VirtualBox/VMware	Lab Virtualization

5. Methodology

The project followed a structured vulnerability assessment methodology consisting of:

Phase 1: Information Gathering

- Identified active hosts.
- Determined IP addresses.

- Mapped network assets.

Phase 2: Network Scanning

- Performed host discovery.
- Conducted TCP port scanning.
- Enumerated services and versions.

Phase 3: Vulnerability Assessment

- Executed OpenVAS scans.
- Identified known vulnerabilities.
- Collected vulnerability details.

Phase 4: Risk Analysis

- Evaluated vulnerability severity.
- Assigned CVSS scores.
- Prioritized remediation efforts.

Phase 5: Reporting

- Documented findings.
 - Provided remediation recommendations.
 - Prepared final assessment report.
-

6. Week-wise Implementation

Week 1: Lab Setup and Initial Learning

Activities Performed

- Installed VirtualBox/VMware.
- Installed Kali Linux.
- Deployed Metasploitable 2.
- Deployed OWASP Juice Shop.
- Configured virtual networking.
- Learned Nmap scanning techniques.
- Configured OpenVAS.

Outcome

Successfully created a controlled testing environment and gained familiarity with vulnerability assessment tools.

Week 2: Network Scanning and Vulnerability Assessment

Network Discovery

Example Command:

```
nmap -sn 192.168.1.0/24
```

Port Scanning

Example Command:

```
nmap -sV -A 192.168.119.130
```

Vulnerability Scanning

Performed comprehensive OpenVAS scans against:

- Metasploitable 2
- OWASP Juice Shop

Outcome

Identified multiple vulnerabilities, exposed services, and misconfigurations.

Week 3: Risk Analysis and Remediation Planning

Activities Performed

- Reviewed scan results.
- Researched security advisories.
- Evaluated CVSS scores.
- Prioritized vulnerabilities.
- Developed remediation recommendations.

Outcome

Produced actionable security recommendations to reduce organizational risk.

7. Findings and Vulnerabilities Identified

Vulnerability 1: Outdated Service Versions

Description

Several services were running outdated software versions containing publicly known vulnerabilities.

Risk Level

High

Impact

Attackers may exploit known vulnerabilities to gain unauthorized access.

Recommendation

- Upgrade software to the latest supported versions.
 - Implement regular patch management.
-

Vulnerability 2: Open Unnecessary Ports

Description

Multiple unnecessary ports were exposed to the network.

Risk Level

Medium

Impact

Increases attack surface and exposure to unauthorized access.

Recommendation

- Disable unused services.
 - Implement firewall filtering rules.
-

Vulnerability 3: Weak Authentication Mechanisms

Description

Default or weak credentials were detected.

Risk Level

Critical

Impact

Unauthorized access and privilege escalation.

Recommendation

- Enforce strong password policies.
 - Enable multi-factor authentication (MFA).
-

Vulnerability 4: Web Application Security Issues

Description

OWASP Juice Shop demonstrated common web application vulnerabilities.

Examples include:

- SQL Injection
- Cross-Site Scripting (XSS)
- Broken Authentication

Risk Level

High

Impact

Data theft, account compromise, and application manipulation.

Recommendation

- Input validation.
 - Parameterized queries.
 - Secure session management.
 - Security testing during development.
-

8. Risk Prioritization Using CVSS

Vulnerability	CVSS Score	Severity
Weak Authentication	9.8	Critical
SQL Injection	9.1	Critical
Outdated Services	8.2	High

Vulnerability	CVSS Score	Severity
XSS Vulnerabilities	7.5	High
Open Ports	5.5	Medium

9. Security Recommendations

Immediate Actions

- Change default credentials.
- Patch vulnerable systems.
- Remove unnecessary services.
- Restrict network exposure.

Short-Term Actions

- Implement vulnerability management processes.
- Establish patch management procedures.
- Deploy endpoint protection solutions.

Long-Term Actions

- Conduct regular security assessments.
 - Implement Security Information and Event Management (SIEM).
 - Develop incident response procedures.
 - Conduct employee cybersecurity awareness training.
-

10. Project Outcomes

The project successfully demonstrated the process of conducting a vulnerability assessment within a small business environment.

Key achievements include:

- Successful deployment of a cybersecurity lab.
 - Practical experience with Nmap and OpenVAS.
 - Identification of multiple security weaknesses.
 - Risk prioritization using CVSS methodology.
 - Development of realistic remediation recommendations.
 - Improved understanding of vulnerability management processes.
-

11. Skills Gained

Technical Skills

- Vulnerability Assessment
- Network Scanning
- Security Analysis
- Risk Assessment
- OpenVAS
- Nmap
- Linux Administration
- Security Reporting

Professional Skills

- Documentation
 - Analytical Thinking
 - Problem Solving
 - Risk Prioritization
 - Technical Communication
-

12. Conclusion

This project provided hands-on experience in vulnerability assessment and risk management by simulating a real-world small business network environment. Through the use of Nmap and OpenVAS, multiple vulnerabilities were identified, analyzed, and prioritized based on risk severity.

The assessment demonstrated how proactive vulnerability management can significantly reduce security risks and strengthen an organization's cybersecurity posture. The project also enhanced practical cybersecurity skills that are directly applicable to Security Operations Center (SOC), Vulnerability Assessment and Penetration Testing (VAPT), and Information Security roles.

References

1. Nmap Official Documentation
2. OpenVAS Documentation
3. OWASP Top 10 Security Risks
4. CVSS v3.1 Specification
5. Metasploitable Documentation
6. OWASP Juice Shop Project Documentation

7. Greenbone Security Documentation
